

"Mine Tycoon"

Created by

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"Mine Tycoon"

1. Introduction

Mine Tycoon is a 2D automation and resource management game developed using JavaFX. The objective is to construct an efficient mining factory, manage resource processing lines, and maximize profits. The game operates on a grid-based placement mechanic where players strategically purchase and position various machines such as Droppers, Conveyor belts, Upgraders, and Furnaces to process raw materials into highly valuable items before depositing them into the Player Bank.

2. Gameplay

The main menu of "Mine Tycoon" is designed to be simple and easy to use. It features only two main buttons:

- **Start:** Click this button to immediately enter the game and start building your factory.
- **Exit:** Click this button to close the game and exit the application.

When the player clicks the "Start" button, they are taken directly to the grid-based factory management screen. Here, players can immediately start placing machines and mining resources to make money without any delay.

(Figure 1) can be used as a reference to see a visual example of the main menu layout.



Figure 1 : The main menu



Figure 2 : The main Game Screen

The Game Screen

- **Grid System:** The core working area where players interact with the environment to place or remove machinery.
- **HUD (Heads-Up Display):** Located on the screen to display the Player Bank's current currency and provide access to the Shop.
- **Shop Manager:** Allows players to spend currency to buy various types of machines. (See Figure 3)
- **Inventory:** A storage area that displays the machines the player currently owns and has available to place onto the grid.
(See Figure 4)

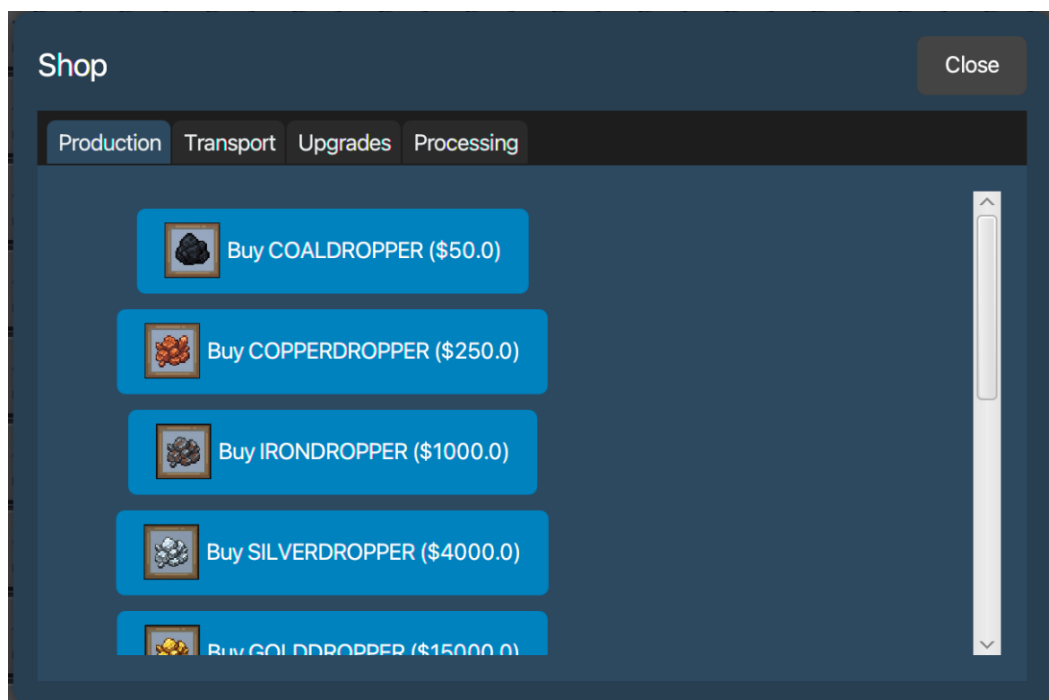


Figure 3 : Shop



Figure 4 : Inventory

Machinery and Items

Players interact with the HUD to deploy these sprites.

- **Droppers:** Generate raw materials over time (e.g., Coal, Copper, Diamond, Platinum). (See Figure 5.1)
- **Conveyors:** Transport items across the grid towards their destination. (See Figure 5.2)
- **Upgraders:** Enhance the monetary value of items as they pass through. (See Figure 5.3)
- **Furnaces:** Melt raw ores and immediately sell the refined items for currency. (See Figure 5.4)

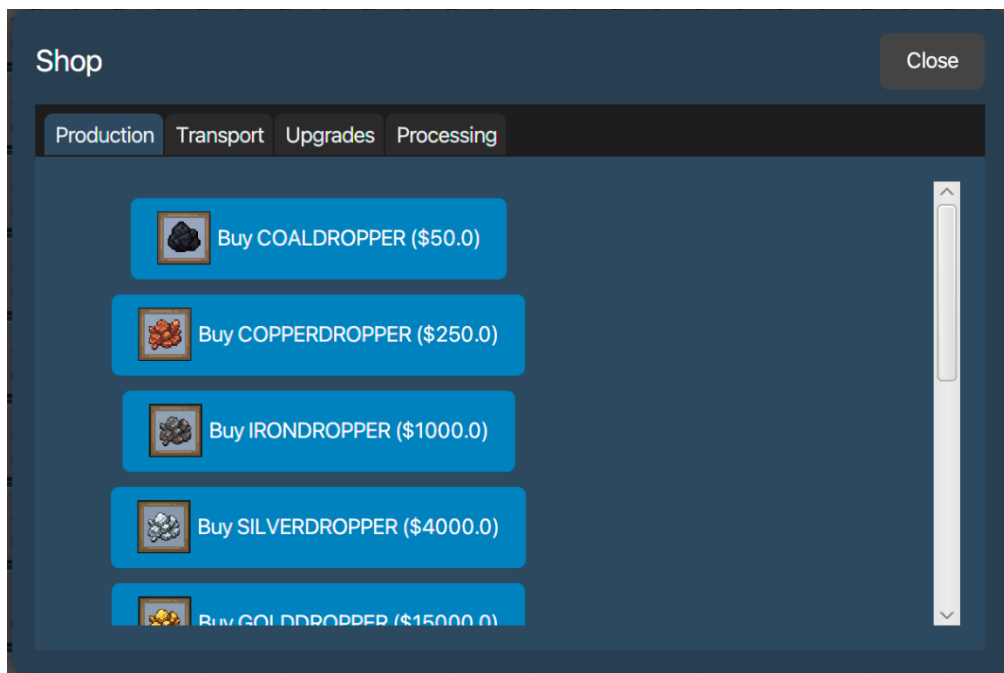


Figure 5.1 : Dropper Shop

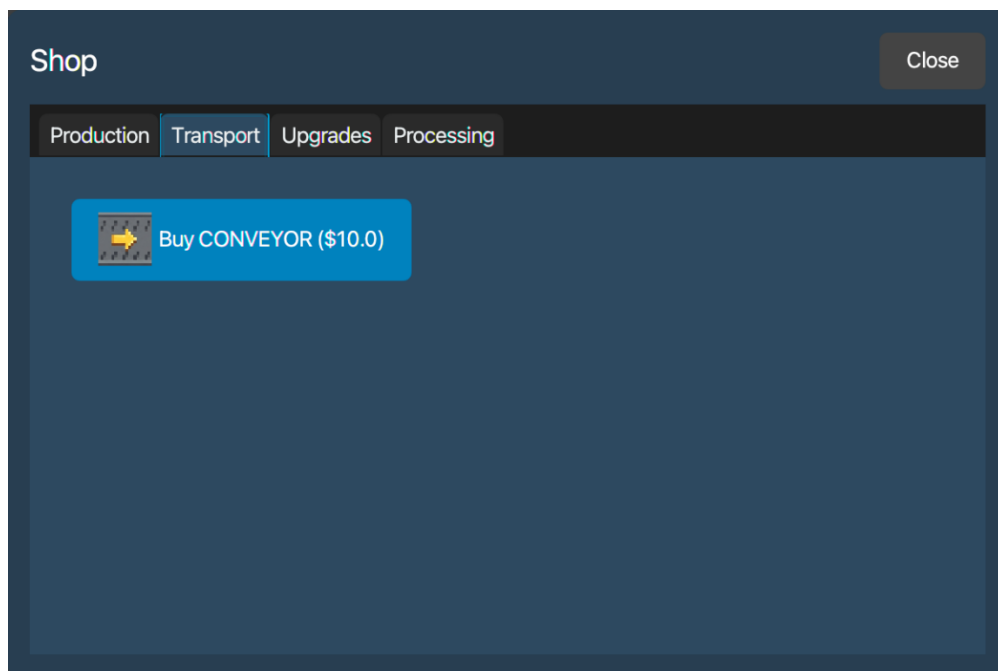


Figure 5.2 : Transport Shop

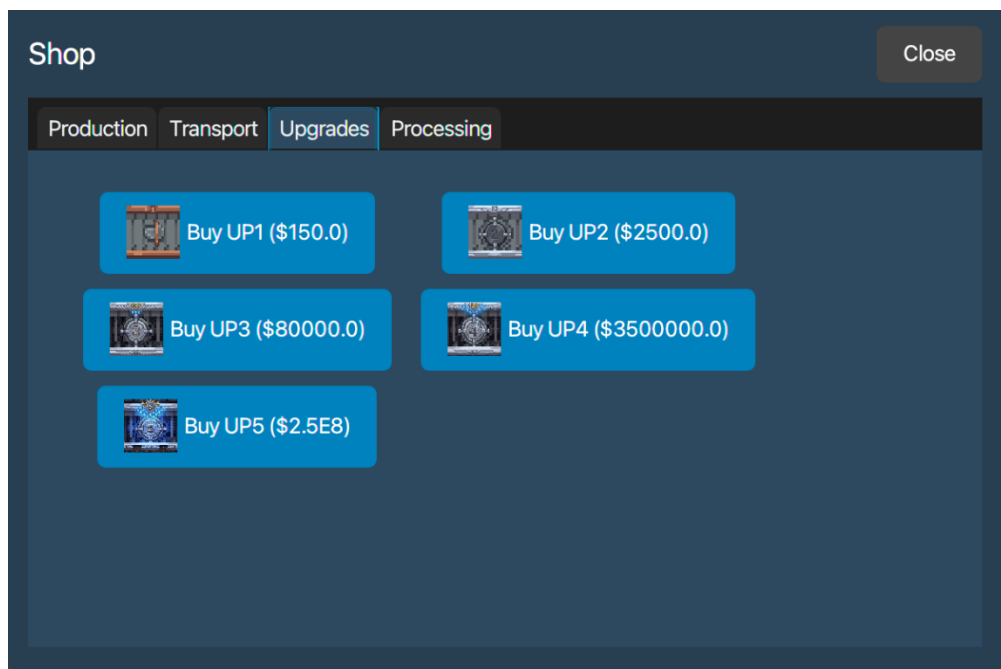


Figure 5.3 : Upgrades Shop

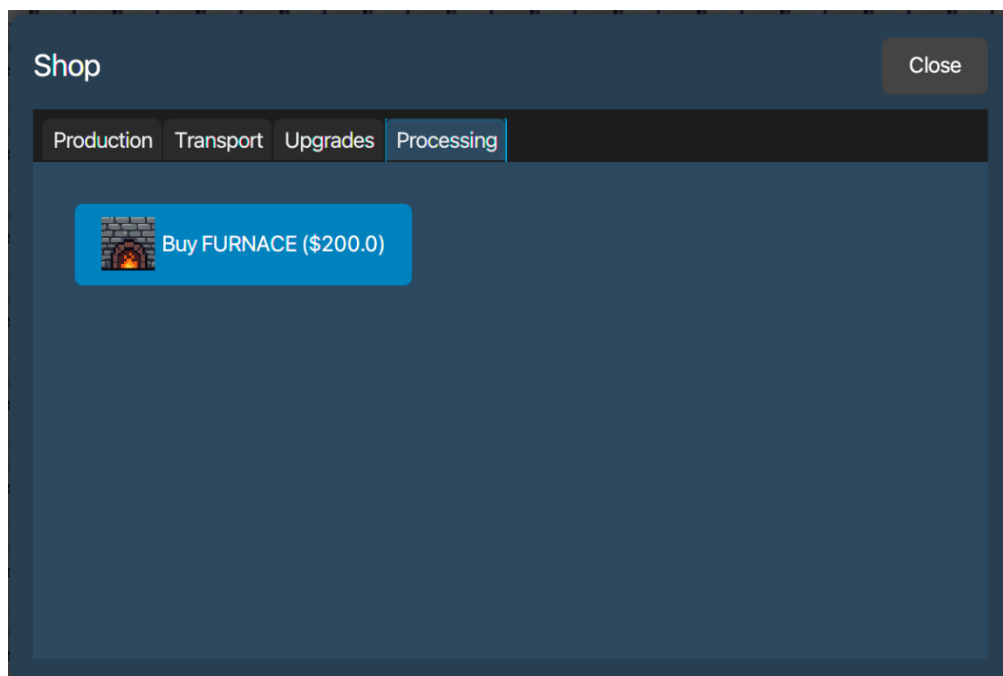


Figure 5.4 : Processing Shop

How to Play

Build an automated factory, process raw materials, and maximize your profits! Basic Controls & Steps:

1. **Buy Machines:** Open the shop menu and spend your starting money to buy a Dropper. Droppers will automatically spawn raw materials (like Coal or Copper) over time.
2. **Move Items:** Place Conveyor Belts to transport your materials across the grid.
3. **Upgrade Value:** Route your items through Upgraders and Furnaces to refine them. Processed items are worth much more money!
4. **Earn Money:** Guide your finished products into the Player Bank to earn currency.
5. **Expand:** Use your profits to buy more expensive and efficient machines to build the ultimate factory!

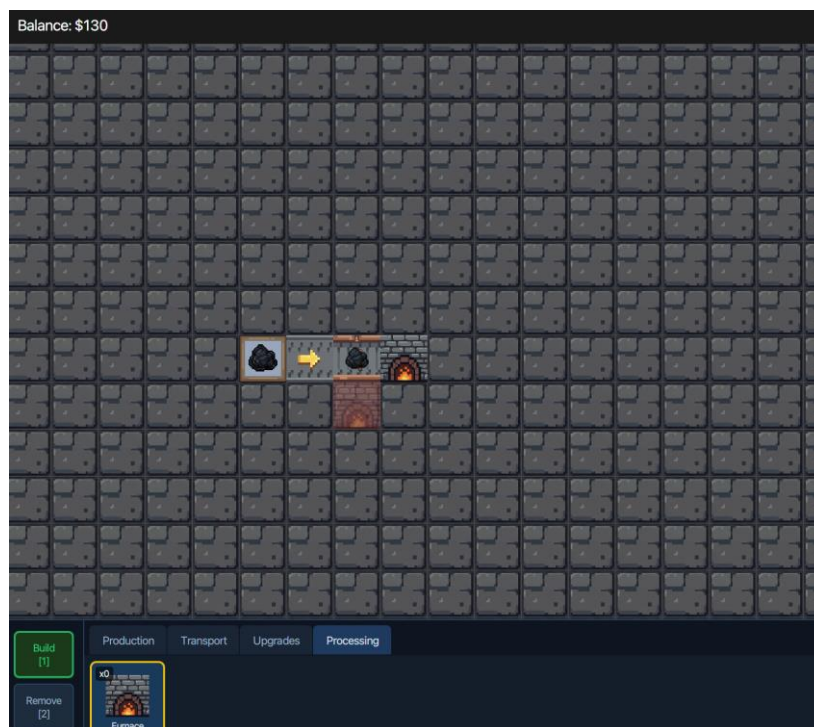
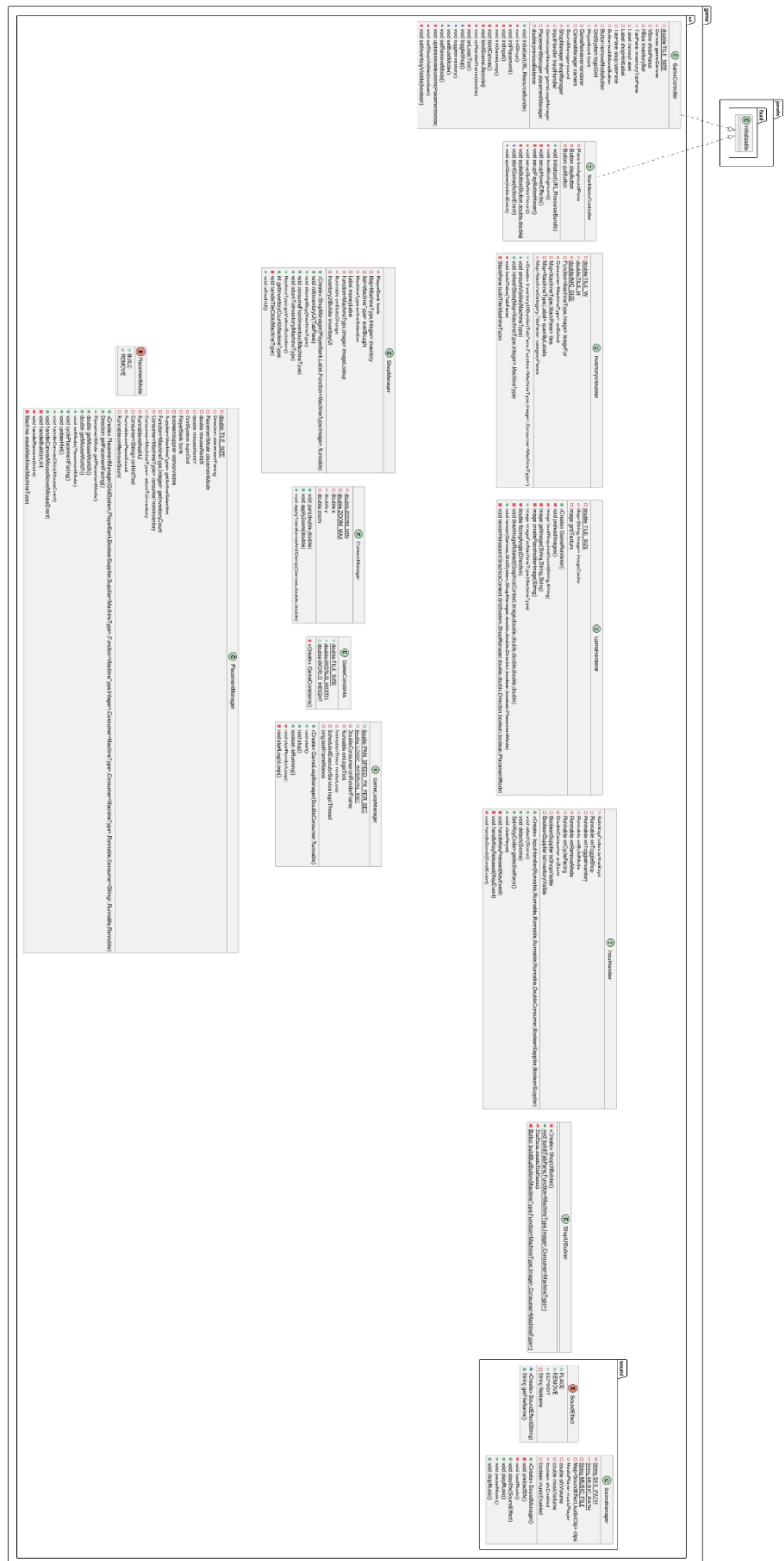
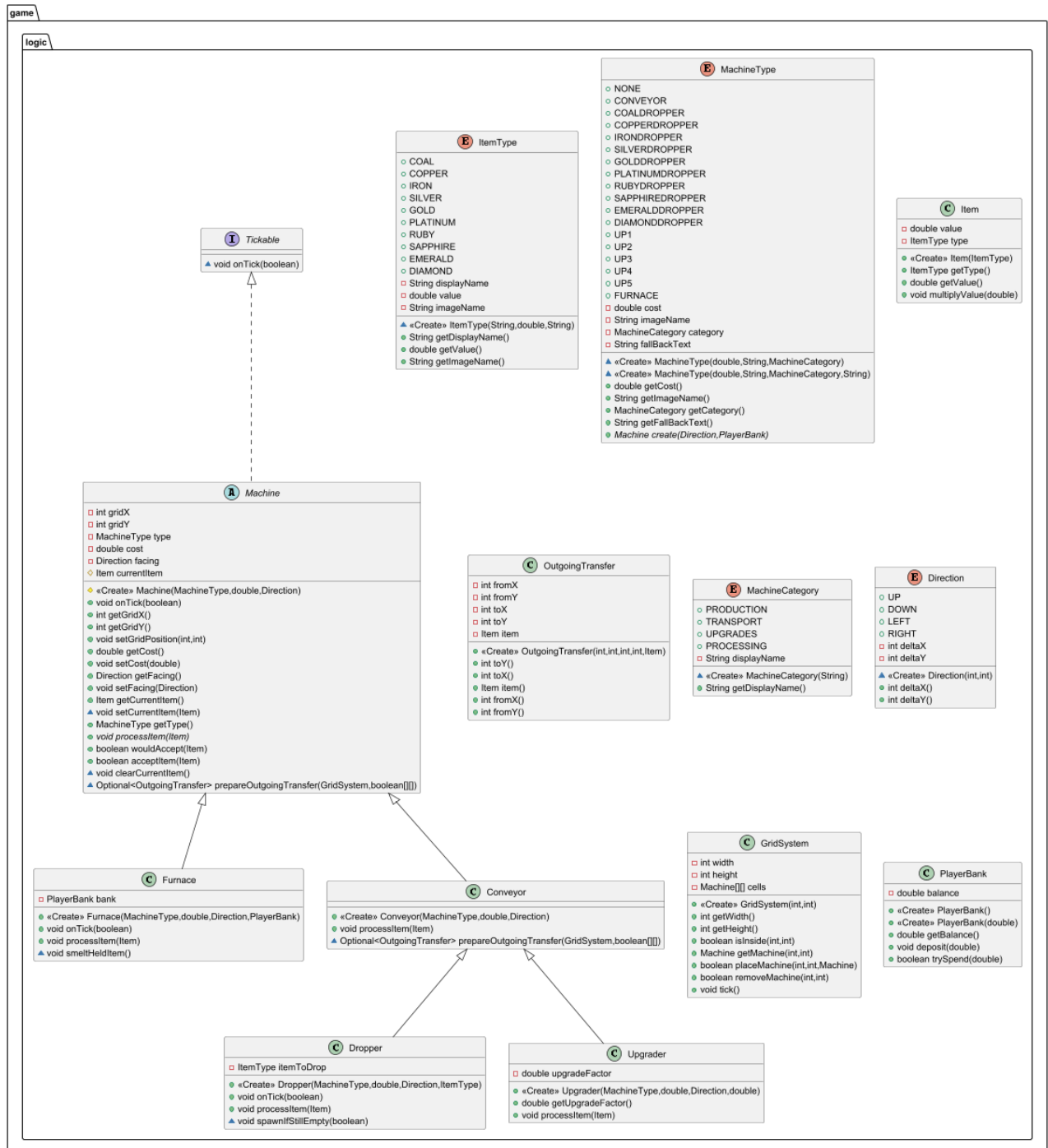


Figure 6 : Basic Factory Setup

3. UML Diagram



3. UML Diagram (continued)



4. Implement Details

<https://itzshen.github.io/MineTycoon-implement-details/>